

Google Map-Based Password Authentication Systems Using Tolerant Distance and Homomorphic Encryption

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Abstract—Passwords are widely used for authentication in Internet applications. Recently, users tend to adopt graphical passwords instead of traditional alphanumeric passwords, since it is much easier for humans to remember images than verbal representations. However, the existing graphical password authentication systems generally suffer from three main issues. 1) It is required to remember and perform complicated operations during the registration/login phases, which significantly limits the systems' usability; 2) The users' passwords are simply stored as plaintexts in servers, and thus the security is compromised; 3) The users need to register/login to each server separately when they are applied in multi-server environment. To address the above issues, we propose a user-friendly and secure Google map-based graphical password (FS-GMGP) system using tolerant distance and homomorphic encryption. By using a homomorphic encryption scheme, each user encrypts his password point and response point selected on Google map, while the servers compute and decrypt the distance between the two encrypted points and then compare the resulting value with a tolerant distance for authentication. Moreover, the FS-GMGP system is extended for multi-server environment. The evaluation results and security analysis show that the FS-GMGP and its extended version achieve desirable usability and security in

single-server environment and multi-server environment, respectively.

Index Terms—Authentication, graphical password, homomorphic encryption, digital forensics, information security, network security.

I. INTRODUCTION

TO MANAGE users' identities and control the access of privacy data, password-based authentication has been popularly adopted in many Internet applications. In the past decades, the traditional authentication systems mostly adopt alphanumeric strings composed of numbers, symbols, and upper- and lower-case letters as passwords. Long random alphanumeric strings are usually used as strong passwords to resist brute force attacks or dictionary attacks, but they are hard for users to remember due to the long-term memory limitations of the human brain as illustrated in [1], [2], [3], [4], [5], [6], [7].

As humans have much higher ability to remember images than verbal representations [1], [2], [3], [4], [5], [6], [7], users can set up a strong graphical password and are able to recall it even after a long period of time. To address the issue of alphanumeric passwords, a variety of graphical password-based authentication systems have been proposed. According to the types of passwords, the existing graphical password systems can be roughly categorized into three classes, i.e., recognition-based [1], [8], [9], [10], [11], [12], [13], redrawing-based [14], [15], [16], [17], [18], [19], [20], [21], and reselection-based password systems [2], [3], [4], [5], [6], [7], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31].

In the recognition-based systems [1], [8], [9], [10], [11], [12], [13], each user is allowed to pass the authentication by recognizing a set of objects displayed on a screen. According to the study in [13], the percentages of users who succeed in the authentication using the recognition-based graphical passwords are much higher than those using the traditional alphanumeric passwords. Consequently, these graphical passwords generally have the advantages in usability and memorability. However, since the number of objects displayed on a screen is usually limited and thus the password space is small, the security of these schemes is difficult to be guaranteed. The redrawing-based systems [14], [15], [16], [17], [18], [19], [20], [21] ask each user to re-draw a special pattern using a finger or mouse. The

Received 1 January 2024; revised 4 March 2025; accepted 4 March 2025. Date of publication 10 March 2025; date of current version 4 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62372125, Grant 62394330, Grant 62471064, Grant 72374058, Grant 62372120, and Grant 62102108, in part by the Guangdong Natural Science Funds for Distinguished Young Scholar under Grant 2023B1515020041, in part by National Science and Technology Council under Grant 112-2221-E-259-007-MY2, in part by the Opening Project of Guangdong Province Key Laboratory of Information Security Technology under Grant 2023B1212060026, in part by Liaoning Collaboration Innovation Center For CSLE, and in part by project SERICS (PE00000014) under the NRRP MUR program funded by the EU - NGEU. (Corresponding author: Zhili Zhou.)

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Digital Object Identifier 10.1109/TDSC.2025.3549028

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